HEADSTART: Forecasting health-system crises from published evidence when local outcome data are scarce

1. Summary

At the start of a health crisis, many useful inputs are already available. Weather and air-quality measurements, calendar effects, wastewater and public surveillance data can often be obtained from public feeds or established partners. The main gap is not usually a lack of these contextual data. It is the small number of reliable outcomes from the specific episode and local service that need to be forecast. Historical local records may be extensive, but they cannot immediately show how a new episode affects cause-specific emergency demand. The model must estimate and update that relationship as episode outcomes arrive.

HEADSTART asks whether published evidence from other settings and past events can improve forecasts before enough outcomes from the current episode are available, and whether the method can detect when that evidence is misleading. Published studies may inform plausible parameter values, useful explanatory variables and lags, candidate model forms, and outcome or warning thresholds. A prior is a probability distribution for model parameters before it is updated with outcomes from the current episode. It represents uncertainty rather than treating a published estimate as exact. HEADSTART will extract this information, measure extraction error and assess how relevant each finding is to Geneva. The main comparison is between a reference strategy and an evidence-informed strategy. Both use the same available context feeds, historical local outcome data and outcomes observed so far in the current episode. Only the evidence-informed strategy uses information extracted from published studies to shape its parameter distributions and model choices.

The work has four linked parts. I will benchmark automated extraction against an independently adjudicated sample and measure how extraction errors affect the uncertainty carried into models. I will build a model that represents pressure on the health system in ordered levels, from routine to critical, while forecasting demand. I will then replay past episodes using only the information available at each forecast date and compare the two strategies. Finally, I will assess whether differences in forecast quality would change operational decisions. Both strategies receive the same weather, air-quality, calendar and surveillance feeds, and both use the same local outcome history. Candidate models are selected under the same rules, set before forecast performance is examined.

The real-world tests are in Geneva and cover three crisis types with different drivers. Respiratory epidemics provide the primary test because published evidence is comparatively rich. Heat-related demand is tested second, and only if the respiratory test succeeds. Legionellosis is a year-four extension using a linked case-environment dataset on which I already lead the data work. It tests case-incidence forecasting and sits outside the emergency-demand comparison. Before clinical data are analysed, the methods will be tested on open Swiss series and released as a public benchmark. If published evidence improves forecasts, the project will show when and by how much. If it has no benefit or causes harm, the result will define where it should not be used. For validated demand-based applications that services authorise, forecasts will run in a daily observation-mode dashboard and be scored against outcomes as they arrive. The dashboard will not direct clinical care.

My background combines fifteen years in quantitative finance, where I worked on rare-event risk, regime changes and stress testing, with a doctorate in biomedical sciences at Geneva. I conceived and built LiteRev with Aziza Merzouki in Olivia Keiser's group, then extended that work into LiteRev-Evidence, the prototype for structured evidence extraction and model comparison. Through GESICA, I

also work with Geneva emergency and surveillance data. HEADSTART builds on these methods and data but tests a distinct question through a programme I will lead.

2. Proposed research

2.1 Current state of research in the field

2.1.1 Forecasts need outcome history as well as context data

Syndromic surveillance detects departures from expected baselines [Farrington 1996; Noufaily 2013], and ensembles of forecasting models perform strongly in data-rich settings [Cramer 2022; Sherratt 2023]. Emergency dispatch records move ahead of confirmed surveillance. The autumn onset of the 2009 A(H1N1) pandemic was identified eight days early in three European regions [Rosenkötter 2013], long series of emergency calls track influenza-like illness [Bonora 2025], and our systematic review maps the field [Edjinedja 2026].

The limitation is not usually a lack of contextual inputs. Weather and air-quality measurements, calendar effects, wastewater and some surveillance indicators may be available in near real time through public feeds or partners, and they can be linked automatically to a forecasting model. What is missing at onset is enough reliable outcome data from the particular episode to estimate how those conditions are affecting this service now. Historical records help establish baseline patterns, but the new episode may differ in its drivers, affected population, severity or service response. The relationship between the available inputs and the local outcome must therefore be checked as new outcomes arrive.

Google Flu Trends illustrates why indirect indicators cannot replace outcome validation. It overestimated influenza by more than a factor of two [Lazer 2014]. Published findings may help, provided their uncertainty and relevance to the local setting are assessed.

2.1.2 Published studies can inform forecasts if uncertainty and relevance are assessed

Statistical models can use earlier studies to inform estimates for a new setting through priors. Methods include power priors [Ibrahim 2000], commensurate priors [Hobbs 2011] and robust meta-analytic-predictive priors [Schmidli 2014]. Informative priors have also been used to forecast disease with little local data [Cook 2023]. The unresolved question is whether an automated pipeline can extract published evidence, preserve its uncertainty, assess its relevance to the new setting and improve forecasts of local health-system demand at crisis onset.

Published studies can inform more than parameter values. They can suggest which variables and time lags to consider, which model forms are suitable for a given crisis and forecast horizon, and how outcomes or warning thresholds have been defined. Living systematic reviews keep evidence current [Elliott 2014]. To my knowledge, however, these different forms of evidence have not been tested together in an operational forecasting workflow under strict historical information constraints.

Automated extraction makes large-scale evidence synthesis possible, but extraction of numerical results is less reliable than extraction of categories (reported accuracy roughly 47–88% versus 74–96%), and omissions account for a large share of errors [Shankar 2026]. HEADSTART will test whether these errors systematically understate uncertainty and whether an explicit measurement-error model can correct the resulting distributions.

Even when a published estimate is extracted correctly, it may not transfer to Geneva because of differences in population, case definitions, service organisation or policy. Formal methods for assessing transportability exist [Bareinboim 2016; Dahabreh 2020; Degtiar 2023], but they have not been

integrated with automated evidence extraction and tested for their effect on forecasts of health-system demand at crisis onset.

2.1.3 Escalation should be assessed as well as the demand count

Emergency services need to know not only how many calls or presentations to expect, but also whether demand is approaching a level that requires action. Surveillance research has used hidden Markov models to distinguish epidemic from non-epidemic periods [Le Strat 1999; Watkins 2009], while extreme-value methods describe rare high-demand events [Coles 2001], including hospital congestion in Switzerland [Ranjbar 2022]. HEADSTART uses an ordered latent state (routine, elevated, strained, critical) as a common way to translate forecasts into levels of system pressure. Mechanistic transmission models [Keeling 2008] are considered only for infectious crises, and heat-related demand is modelled through exposure and response. Recent changes in variance and lag-one autocorrelation, which rise before some transitions [Scheffer 2009] and have been studied in epidemics [O'Regan 2013; Brett 2018; Southall 2021], are also tested as possible additional indicators of an approaching transition, not assumed to be reliable early-warning signals.

2.1.4 Forecast quality matters when it can change a decision

A forecast is useful when it changes a decision in a defensible way. Proper scoring rules evaluate probabilistic forecasts, including their calibration [Gneiting 2007], while decision-analytic methods assess predictions in light of consequences [Vickers 2006]. Heat thresholds based on mortality need not match thresholds for service demand [Lung 2021], and reported pollution–asthma emergency associations vary across settings [Zheng 2015]. HEADSTART will therefore evaluate forecasts first statistically, then under an operational loss structure elicited from emergency-service experts.

2.1.5 The gap: testing published evidence in a new setting

Current programmes include multi-model forecasting hubs such as ECDC RespiCast and US CDC FluSight, the Horizon Europe epidemic-intelligence project GeoAI4EI, and the Franco-Swiss GESICA programme. The question here is narrower. Does automatically extracted evidence from earlier studies improve forecasts of a local service's demand when observations from the current episode are still few? HEADSTART tests this under registered rules, with the same data streams for both strategies and a historical reconstruction that excludes later information.

2.2 Current state of personal research and competences required for the project

My background combines fifteen years in quantitative finance, where I worked on risk, extreme values, regime models and stress testing, with a doctorate in biomedical sciences at Geneva (defended 18 December 2023). This combination provides the methods for modelling rare events and the public-health training needed to apply them to emergency services.

2.2.1 Prediction under sparse and imperfect information

In [Orel 2022] I predicted individual HIV status from socio-behavioural characteristics across East and Southern Africa and examined where models transferred between countries. [Merzouki 2021] and [Esra 2023] developed related work. As last author of [Ng'ambi 2026] (accepted), on cardiovascular-disease classification across harmonised WHO STEPS surveys, I have taken a more senior role in this line of research. It provides direct experience with the question of when model relationships transfer between settings.

2.2.2 Automated evidence synthesis

I conceived and built LiteRev [Orel 2023] with Aziza Merzouki (PhD, computer science), in Prof. Olivia Keiser's group. The tool searches open databases using natural-language or Boolean queries, removes

duplicates, maps the available literature and suggests relevant papers in repeated rounds. I took the initiative to secure CHF 30,000 from UNIGE, CHF 10,000 from Venture Kick and CHF 20,000 from Mimosa, outside a group grant. The systematic review of AI in emergency medical services [Edjinedja 2026], within GESICA, uses LiteRev in its methods.

2.2.3 Outbreak and health-system modelling in Switzerland

In [Orel 2024] I compared clinical severity between Delta and Omicron sub-lineages in a Swiss tertiary centre. [Estill 2020] developed age-structured scenarios for the Swiss SARS-CoV-2 epidemic to support decisions under time pressure. That work helped motivate the present question. I also contributed to WHO African Region reporting and seroprevalence estimation [Nwosu 2021].

2.2.4 The prototype and test bed, LiteRev-Evidence

Since 2024, I have extended LiteRev into LiteRev-Evidence, a working prototype for structured numerical extraction and modelling. It searches and deduplicates the literature, records screening decisions with PRISMA accounting, extracts study details and PICO elements (population, intervention, comparator and outcome), tracks the source and certainty of each extracted number, and pools results into distributions of plausible parameter values. It then suggests explanatory variables and their supporting studies, identifies candidate outcomes, fits models across algorithm families and evaluates them with time-series validation that prevents future observations from entering model selection. It also builds compartmental model components from published parameter ranges. Evidence updates are reviewed before they are applied. The prototype already contains the main steps to validate.

For GESICA I built the data foundation for the Geneva–Vaud–Neuchâtel network, whose Geneva component HEADSTART uses. It covers 77 notifiable diseases in eight model classes and an inventory of 28 surveillance sources (23 infectious and 5 environmental). Each source is documented for coverage, resolution, latency, access and quality.

The prototype reduces development risk, since its literature-search, extraction and model-selection components already run in preliminary form. The grant focuses on the scientific questions that remain open, namely whether the extracted evidence is sufficiently reliable, whether it transfers to Geneva and whether it improves forecasts under the registered comparison.

2.2.5 Linked data on a contrasting crisis type

I lead the data work for a cantonal study already under way (BASEC 2026-00324, ethics granted) that links confirmed legionellosis cases in Geneva to individual domestic hot-water installations. These linked data provide a contrasting setting for the year-four case-incidence extension.

My quantitative-finance work provides experience in Bayesian hierarchical modelling, regime changes, extreme values and stress testing under misspecification. LiteRev-Evidence provides the extraction and pooling methods. My doctorate, Swiss COVID-19 work and GESICA provide the public-health and Geneva context.

2.3 Detailed research plan

2.3.1 Objectives and hypotheses

The central hypothesis is that published evidence can help estimate the relationship between these inputs and the target outcome, provided its uncertainty and relevance to Geneva are represented.

	Role	Statement
H1	Evidence quality	Does automated extraction preserve the uncertainty reported in studies?

	Role	Statement
C2	Model adequacy	Can the model identify levels of pressure and estimate their probabilities reliably?
H3a	Main hypothesis	Does adding published evidence improve early forecasts when outcomes from the current episode are few?
H3b	Safety check	Can the model reduce reliance on published evidence when it conflicts with local outcomes?
H3c	Secondary question	Do short-term changes in variability add useful predictive information?
H4	Decision value	Does any gain change an operational decision for the better?

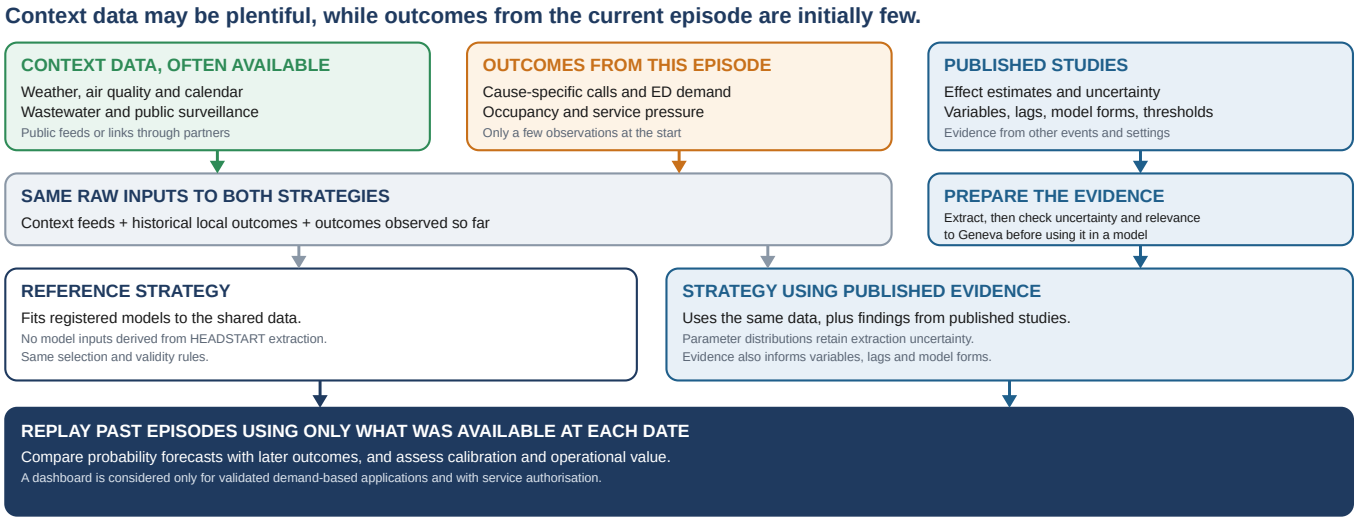


Figure 1. The context data shared by both strategies, evidence from published studies, and evaluation against later outcomes.

O1. Make published evidence usable without hiding its uncertainty

H1. Automated extraction will understate the uncertainty in quantitative evidence, including reported within-study uncertainty and variation between studies, making the resulting parameter distributions too narrow. A measurement-error layer should reduce this under-dispersion. Success will be assessed using pre-specified coverage and calibration measures on held-out studies adjudicated by independent human extractors.

Omissions are common [Shankar 2026], and H1 tests whether they also reduce the estimated spread of the evidence and whether the correction restores useful uncertainty. The primary benchmark concerns parameter values, and the other classes are checked against published syntheses.

O2. Represent escalation in a form that separates the state from the point forecast

C2, model adequacy criterion. The state model must recover interpretable levels of pressure and produce calibrated probabilities at a comparable false-alarm rate. This shared state scale allows forecasts from different candidate models to be compared in operational terms. It does not assume that a regime-switching model will outperform simpler threshold-based methods.

Simulation tests whether the states and their transitions can be identified from the available data channels. Forecast calibration provides a second check. If the state model fails either criterion, the registered ordinal state-space fallback is used, and the main comparison of the two strategies still proceeds.

O3. Test whether published evidence improves forecasts and identify when it harms

H3a. During the early days of an episode, a strategy informed by published studies (parameter distributions, explanatory variables and lags, and candidate model forms) will produce better probabilistic forecasts than a reference strategy that uses the same context inputs, local records and outcomes observed so far, but not the literature-derived model inputs. How long any advantage persists as outcomes accumulate will be assessed secondarily.

H3b. When the published evidence is consistent with the process generating local outcomes, adaptive weighting, which reduces the weight given to that evidence when local results conflict with it, should perform no worse than fixed weighting, within the pre-specified margin Δ. When evidence is deliberately made inaccurate, adaptive weighting should outperform fixed weighting. This tests whether the model can limit harm rather than assume published evidence is always helpful.

Δ is set to half the minimum improvement defined as meaningful under H3a. Under the registered simulations where published and outcome-generating distributions match, this 50% preservation rule allows adaptive weighting to give up no more than half of that pre-specified benefit.

H3c. Short-window changes in variability and lag-one autocorrelation will be tested for whether they improve forecasts beyond the context inputs, the local level and trend, and the published evidence.

Primary confirmatory comparison. The evidence-informed and reference strategies use the same pre-registered selection procedure and are compared with the continuous ranked probability score (CRPS), which evaluates full probability forecasts. The primary test uses respiratory episodes and the pre-specified early-episode window. Heat is tested with the same comparison only if the respiratory test passes, and this fixed order controls the family-wise error rate. How the difference changes as more local outcomes arrive is a secondary analysis.

O4. Establish whether predictive improvement matters for decisions

H4. For the demand-based settings, the forecast with the best statistical score will not always be the best operational choice. Decision analysis will assess the models under losses and escalation thresholds elicited from emergency-service experts. An evidence-informed forecast is useful only if its gain changes a decision in a way that matters operationally.

Scope. Three Geneva crisis types, fixed for the grant

HEADSTART covers three Geneva crisis types. (1) Respiratory epidemics, including SARS-CoV-2, influenza and RSV, provide the primary confirmatory test because the published evidence base is comparatively rich. (2) Heat events provide the sequential generalisation test, with ozone and PM10 treated as co-exposures or effect modifiers rather than separate domains. (3) Legionellosis provides a year-four case-incidence extension based on linked case-environment data. No fourth operational domain is planned. For respiratory and heat, both strategies forecast the same target, daily emergency demand from CASU-144, restricted to cause groups specified in advance. For respiratory epidemics, the target covers respiratory-related demand. For heat, it covers heat-sensitive demand, with dehydration, renal and psychiatric causes specified in advance, while respiratory-related demand and demand among people aged 75 or older are sensitivity analyses [Schulte 2024; Ragettli 2019]. Legionellosis forecasts case incidence and is outside the emergency-demand comparison. If CASU-144 fails the month-12 access and quality gate, the registered fallback is activated and the claim narrows.

What the project does not claim

The project is not intended to outperform established forecasting hubs when long outcome histories are available. It does not assume that published evidence will improve forecasts in every setting, and it

does not promise clinical decision-making by an automated system. The dashboard, where authorised, will operate in observation mode only.

2.3.2 Work packages and methods

WP1. Turn published evidence into model inputs (M1–M20)

T1.1. Define the evidence target and register the protocol (M1–M4)

The protocol covers four classes of evidence and is fixed before extraction. (1) Parameter values, the quantitative core, include weather-demand effects (temperature, heat-day exceedance, ozone and PM10), surge magnitude and timing, length-of-stay and occupancy distributions, and admission fractions. Transmission parameters and shedding-to-incidence conversions are secondary. (2) Explanatory variables and their lags reported as informative for each crisis type. (3) Model-form evidence on which algorithm families perform for which outcome and horizon [Wargon 2009]. (4) Outcome definitions and warning thresholds [Lung 2021]. These evidence classes inform candidate model inputs and the later interpretation of forecasts. The full extraction benchmark focuses on class 1, while classes 2–4 carry provenance and certainty grades and are validated against published syntheses.

T1.2. Build the quantitative extraction benchmark (M3–M9)

A separately contracted independent extractor and I will each extract target quantities from a stratified random sample of 300 publications, followed by adjudication. The sample will cover parameter classes, reporting quality, study design and source type, including journal articles and timestamped situational reports. This allows error to be assessed separately for sources used in secondary analyses. An external benchmark will compare retrieval with inclusion sets from published systematic reviews and extracted and pooled values with their published summary estimates. The benchmark will be released openly.

T1.3. Characterise automated extraction error (M7–M14)

Automated extraction will be compared with the adjudicated benchmark on point estimates, uncertainty, omissions and variation between studies. H1's primary metric is the median log-ratio of the standard deviation from automated extraction to that from the adjudicated evidence distribution, by parameter class on a common scale and under the registered pooling model. A ratio below one indicates under-dispersion. Within-study uncertainty and between-study heterogeneity are propagated separately and jointly, with sensitivity analyses for the extraction model version.

T1.4. Construct uncertainty-aware evidence distributions (M12–M18)

Extraction errors are represented as measurement error. The task compares inverse-variance and quality-weighted pooling, meta-analytic-predictive priors and power-prior methods. Each resulting parameter distribution records its sources, extraction uncertainty and the weight assigned to external evidence.

T1.5. Transportability screen (M15–M20)

This task assesses effect modifiers and differences between study settings that may affect relevance to Geneva. Criteria for transportability, and the mapping from the resulting score to prior widening or reduced evidence weight, are fixed before the confirmatory forecast evaluation.

Deliverables are D1.1 (open extraction benchmark), D1.2 (extraction-error analysis) and D1.3 (evidence-derived model-input library).

Risk and mitigation. If expert time is limited, the benchmark narrows to the core parameter classes.

WP2. Model levels of health-system pressure (M1–M28)

T2.1. Specify and identify the state model (M1–M9)

The model is a Bayesian hierarchical Markov regime-switching model [Hamilton 1989], meaning that it estimates how a system moves between ordered levels of pressure. The default levels are routine, elevated, strained and critical. Cause-filtered emergency calls are the primary observation channel, and ED presentations and ICU occupancy are added where access permits. Weather, air quality, calendar effects and epidemic indicators enter as explanatory variables. Before real data are fitted, simulation tests whether the states and their transitions can be recovered, including a version with emergency calls alone. If recovery is poor, the fallback is a simpler ordinal state-space model. The number of levels and their anchors are fixed at registration. Where the literature provides suitable definitions, these guide the number of levels and the thresholds. Otherwise the default anchors are seasonal percentiles, routine below P75, elevated P75 to P90, strained P90 to P97.5 or under sustained capacity pressure (ICU occupancy above 85% or ED boarding above its seasonal P90), and critical above P97.5 or at capacity saturation. The highest applicable state prevails, and capacity saturation alone defines critical. These anchors inform the model but are not hard cut-offs. Later expert elicitation can refine their decision interpretation, not change the confirmatory state definitions.

T2.2. Represent the critical tail (M6–M14)

Rare high-demand events are modelled with peaks-over-threshold methods and a generalised Pareto distribution, linked to the probability of the critical state. Threshold sensitivity is reported, and this work supports the main comparison rather than defining a separate hypothesis.

T2.3. Include published evidence in the model parameters (M10–M20)

WP1's distributions are mapped to parameters for which published evidence is relevant, including weather effects, surge magnitude and transitions between pressure levels. Three approaches are compared, weakly informative reference distributions, fixed use of evidence-derived distributions, and adaptive weighting that reduces the evidence weight when the registered prior-data conflict measure signals disagreement. The conflict measure and weight-reduction rule are fixed before evaluation. Power-prior and commensurate-prior methods are used as sensitivity comparators.

T2.4. Add resilience indicators as a secondary information channel (M12–M20)

Rolling variance and lag-one autocorrelation, possible indicators of changing dynamics, are tested as optional predictors of state transitions. The question is whether they add information beyond context inputs, recent outcome level and trend, and published evidence. A null result remains informative.

T2.5. Calibration, automation and validity monitoring (M20–M28)

The task uses calibration methods suited to time-dependent data and tests conformal prediction [Angelopoulos 2023; Barber 2023] as a robustness check if simulation supports its assumptions. Public context feeds, including weather, air quality, wastewater and surveillance indicators, are ingested automatically at their native update frequency. Partner data are linked only under access agreements. The pipeline records timestamps, data revisions and model versions, refits at the registered cadence, and checks residual structure, dispersion and calibration after each fit. Models failing pre-specified validity checks are excluded. A reference implementation is released in LiteRev-Evidence.

Simulated trajectories are used to test identifiability and model behaviour, not as extra observations. Treating literature-based simulations as real outcomes would count the same information twice and undermine the test for conflict between the prior and local data. Simulations therefore do not tighten the evidence-derived distributions.

Deliverables are D2.1 (model specification and identifiability study), D2.2 (open implementation) and D2.3 (methods paper).

Risk and mitigation. If the states cannot be separated reliably, the registered ordinal model is used and the identifiability limit is reported.

WP3. Test forecasts using historically available information (M12–M42)

T3.0. Outcome hierarchy, data-access gate and episode eligibility (M1–M14)

The outcome hierarchy, access criteria and episode-eligibility rules are registered before forecast performance is examined.

Primary outcome. Daily respiratory-related emergency demand is derived from Geneva CASU-144 records using a pre-registered classification of call reasons and urgency levels, and it is not the raw total number of calls. The heat comparison uses the same series and construction, restricted to cause groups specified in advance (dehydration, renal and psychiatric causes). Respiratory-restricted and age-75-or-older analyses are sensitivity checks. ED presentations and ICU occupancy are additional observations for the shared pressure-state model. Weather, air quality, wastewater, sentinel consultations and calendar effects are context inputs.

Outcome hierarchy and geography. The primary target is cause-filtered CASU-144 emergency demand in the canton of Geneva, at daily resolution. If it fails the month-12 access and quality gate, the registered hierarchy is activated, ED presentations by category, then ICU occupancy, then all-cause 144 engagements or hospital admissions for sensitivity analyses. Open series such as Sentinella and wastewater can support methodological validation but do not by themselves substitute for a successful emergency-demand test. This is a deliberate single-canton design, and transfer is tested across the three crisis types and on the open national benchmark.

Data-access gate. CASU-144 remains the pre-specified primary outcome. Criteria for historical depth, resolution, reporting delay and completeness are fixed in advance. At month 12, the gate determines whether the series meets them. If not, the registered fallback is activated and the claim is narrowed, and the fallback is not selected based on model performance.

Episode eligibility. An episode enters the confirmatory comparison only if the onset can be identified using the prospective onset rule, enough pre-onset history is available for the baseline, enough post-onset outcomes exist at the primary forecast horizon, the literature available at each historical origin can be reconstructed, future information can be excluded, and the episode can be separated from adjacent episodes. Ineligible episodes remain for sensitivity analyses. Co-circulating respiratory pathogens count as one demand episode.

Registration. The outcome hierarchy, access gate and eligibility rules are registered first. After the month-12 checkpoint and inventory of eligible episodes, but before forecast performance is examined, a second registration fixes the early-episode window N, the primary horizon for each crisis type and the non-inferiority margin Δ. Horizon choice follows a pre-specified operational criterion, not observed performance or episode-specific results.

Open-data validation track (M12–M24, secondary and pre-specified). Before clinical records are connected, the full evaluation pipeline runs on open Swiss series (federal respiratory surveillance, national wastewater data, weekly all-cause deaths and MeteoSwiss weather data). Archived multi-model forecasts provide contemporary comparators [Cramer 2022; Sherratt 2023]. The benchmark checks whether the historical-reconstruction and model-selection machinery works as intended. It also shows which context feeds can be linked from public sources, while the Geneva clinical data are needed to test the primary emergency-demand question.

T3.1. Assemble the retrospective information set (M12–M20)

The CASU-144 series is harmonised with available context feeds and other outcome channels. Completeness and reporting delay are quantified, and right-truncation is addressed with nowcasting [Höhle 2014; McGough 2020], because reporting may worsen under pressure and mimic an early-warning signal. HUG COVID-19 hospitalisation data that I previously analysed [Orel 2024] are re-requested as a secondary respiratory validation and calibration channel. Pharmacy sales and wastewater measurements are requested through the cantonal pharmacist and used as syndromic inputs where access and coverage permit.

T3.2. Reconstruct the true information set (M18–M30)

For each historical episode, successive forecast origins use only information available at that date. The confirmatory evidence comes from peer-reviewed papers and preprints published before each origin. Timestamped WHO/ECDC reports on events elsewhere are included only in a registered secondary analysis, which tests a broader evidence feed, not the primary literature-based strategy. Text describing the local episode is excluded from its evidence inputs because it may reveal the outcome being forecast. Each input has an observation timestamp and a release timestamp. When a source is revised, the reconstruction uses the version that was available at the historical origin, not the final corrected value. Automated feeds are replayed at their native cadence. Candidate horizons are 7, 14 and 28 days, and the primary horizon is fixed at the second registration using the operational rule. The early-episode window starts at an onset rule that uses only information available at the time.

T3.3. The model library and the automated selection engine (M20–M34)

The model candidates are registered before evaluation and organised into two strategies. The reference strategy uses statistical, surveillance and machine-learning candidates (a seasonal GLM with weekday and holiday terms, persistence, Farrington and Noufaily exceedance methods, Bayesian nowcasting [Salmon 2016], penalised, gradient-boosted and upper-tail quantile learners, and the WP2 state model with weakly informative distributions) fitted to historical local outcomes, outcomes already observed in the current episode, and available context feeds. It does not use the project's extracted evidence to set parameter distributions or guide scenario-specific variable, lag or model-form choices. The evidence-informed strategy uses the same raw input streams and outcome history, adding WP1's evidence-derived parameter distributions and registered recommendations about explanatory variables, lags and model forms. Its three versions use fixed evidence weights, adaptive weighting based on prior-data conflict, or adaptive weighting plus the secondary resilience indicators. Mechanistic transmission components are allowed only for infectious crises, and environmental crises use exposure-response relationships. Permitted candidates are fixed by crisis type at registration.

Model selection is nested to prevent leakage. At each historical forecast origin, each strategy selects its best eligible candidate using only inner rolling validation on outcomes observed before that date. The CRPS is the primary ranking criterion, while log score and calibration are pre-registered eligibility checks, not additional ranking criteria. The validity checks in T2.5 also apply. The selected model then produces the forecast for the later outer period, whose outcomes were not used in selection. Both strategies use the same selection rules. The main comparison tests the full evidence-informed strategy rather than one isolated algorithm, and secondary contrasts examine the contributions of parameter priors, variable/lag choices and model forms.

Confirmatory testing uses a fixed sequence.

1. Test 1, primary. Compare the evidence-informed strategy with the reference strategy on the episode-level CRPS skill score for respiratory episodes, over the registered early-episode

window N. The test uses α = 0.05, two-sided, because harmful effects are as important as benefits. Success requires a paired-permutation p-value below α and a lower confidence bound above the pre-registered minimum relevant improvement.

2. Test 2, generalisation. If and only if Test 1 passes, repeat the same comparison for heat-sensitive emergency demand. The fixed order controls the family-wise error rate.
3. If Test 1 fails, H3a is not supported and the heat comparison is exploratory.

All other comparisons are secondary. The reference strategy's candidate list and a pre-declared range of weakly informative parameter settings are fixed at registration. This prevents the comparator from being weakened after results are known.

Primary endpoint. For each episode and strategy, CRPS is averaged across eligible forecast origins in the registered window at the primary horizon. The episode-level skill score is 1 − CRPS_evidence / CRPS_reference, so positive values favour the evidence-informed strategy. The primary statistic is the equally weighted mean across episodes. The episode is the unit of inference, since there are only 13–14 candidate respiratory episodes before eligibility screening. A paired permutation test and an episode-level bootstrap are both pre-registered. The bootstrap resamples whole episodes and therefore preserves temporal dependence within each episode, and any disagreement between methods is reported. Simulations conducted before the second registration assess type-I error, power and interval width under realistic dependence. They also set the minimum number of eligible episodes below which no confirmatory claim is made, and choose N as the smallest candidate window meeting the pre-specified operating-characteristic criterion, without using forecast performance. The minimum relevant improvement is fixed at the second registration using an operational-relevance rule defined before outcome evaluation. The later T4.1 elicitation translates the effect into consequences but does not determine that threshold. Δ is half this minimum relevant improvement. The same simulations test the stability of the ratio-based skill score when baseline CRPS varies between episodes. Secondary measures are log score, calibration (PIT and interval coverage), and detection of escalation at matched false-alarm rates. For H3b, adaptive weighting is non-inferior when its CRPS skill score is no more than Δ below fixed weighting in simulations where the evidence and outcome-generating distributions match within the registered tolerance. Under deliberate shifts in location, scale and transfer, adaptive weighting must outperform fixed weighting. Real episodes provide complementary evidence.

T3.4. Map benefit and failure (M28–M38)

The analysis identifies episodes where published evidence improves or worsens forecasts and examines likely reasons, such as differences in population, outcome definition, service organisation, policy or time period, as well as extraction uncertainty and conflict with the local outcomes. Deliberately shifted priors provide stress tests. A key question is whether adaptive weighting reduces harm early enough. Results will show when the approach helps, when it does not, and why, rather than a single average effect.

T3.5. Test generalisation (M34–M42)

The heat test is the sequential generalisation. If time and data permit, the year-four legionellosis extension applies the same principles to case-incidence forecasts at outbreak onset.

Deliverables are D3.1 (reproducible historical forecast-evaluation pipeline), D3.2 (primary result), D3.3 (benefit and failure analysis) and D3.4 (analysis across crisis types).

Risk and mitigation. Access to operational records may be delayed. Agreements are being pursued before the award. The open-data track can validate the reconstruction and evaluation methods without

clinical data, but it cannot replace the main test of local emergency demand. If access fails, the scientific claim narrows and the decision analysis is limited accordingly.

WP4. From predictive skill to operational value (M24–M48)

T4.1. Elicit operational losses and thresholds (M24–M32)

A structured SHELF elicitation (n ≈ 15–20 experts from HUG emergency medicine, CASU-144 regulation and capacity management) will assess the consequences of early, late and unnecessary escalation. Thresholds are derived from these elicited losses. The primary decision is defined in advance. Surge-capacity escalation is triggered when the forecast probability of the strained or critical state exceeds the elicited threshold. For candidates that forecast counts rather than a state, a registered mapping converts the predictive distribution to the same escalation-state probabilities.

T4.2. Decision-analytic evaluation and equity audit (M30–M40)

The WP3 forecasts are assessed with net-benefit and decision-curve analysis and value-of-information methods to determine whether rankings change when operational consequences are included. Calibration and threshold performance are also checked across aggregate groups by age, sex and neighbourhood deprivation. Overall calibration is not enough if the model performs poorly for a decision-relevant group.

T4.3. Counterfactuals, observation-mode dashboard and living updating (M34–M48)

For selected historical episodes, a simple capacity model estimates what might have changed if escalation had followed the forecast signal. Uncertainty is propagated, and the results are not presented as causal estimates. Where a demand-based strategy passes forecast and decision-value checks and the services authorise it, the project will run a daily dashboard in observation mode. It will publish forecasts, pressure-state probabilities and validity checks, record outputs and score them against outcomes as those outcomes arrive, and it will not direct clinical care. Legionellosis remains a retrospective case-incidence extension and does not need to enter the capacity decision or dashboard. The literature monitor checks for new studies daily, while full scenario syntheses are refreshed on a registered schedule [Elliott 2014]. Updates to priors, variables, thresholds or model choice are version-controlled and applied only under pre-registered rules. If authorisation is not granted, the retrospective evaluation and documented update protocol remain deliverables.

Deliverables are D4.1 (loss structure and equity audit), D4.2 (decision-analytic evaluation) and D4.3 (counterfactual analysis and, where authorised, the observation-mode dashboard and update protocol).

Reproducibility and data protection. Code is version-controlled in R and Python under a registered analysis plan. Clinical data are processed in the UNIGE secure environment under applicable institutional and regulatory approvals. The extraction benchmark and software will be released openly, with synthetic equivalents where data cannot be shared.

Expected outputs are four to six papers and reusable resources, including the extraction benchmark, public forecasting benchmark, evidence-derived input library and decision-analysis framework. I will lead the methodological, benchmark and integrative outputs. The results will be reported whether the evidence-informed strategy helps, has no effect or causes harm.

2.3.3 Research environment, team and resources

Host. The project will be based at the Institute of Global Health, Faculty of Medicine, University of Geneva, with methodological collaboration from Data Science for Digital Health (Prof. Douglas Teodoro) and clinical collaboration from HUG emergency medicine (Prof. Thibaut Desmettre and Dr Robert Larribau, CASU-144).

The Institute brings together epidemiology, infectious-disease modelling and evidence synthesis, and the programme will be hosted alongside its groups and led independently. DS4DH provides the biomedical natural-language-processing expertise needed for WP1, and GESICA has already established working links with emergency-medicine partners [Edjinedja 2026]. Existing infrastructure includes LiteRev-Evidence, the linked legionellosis study, UNIGE HPC and the University's secure clinical-data environment.

Commitments, by status.

Item	Status	Evidence or action
Legionellosis linked data	**Secured**	BASEC 2026-00324, data via the médecin cantonal, SIG technical data
Host institute (ISG)	**Agreed**	Hosting accepted, SNSF-template signatures in preparation
Computing	**Secured**	UNIGE HPC and secure-environment access in place
DS4DH methodological collaboration	**Secured**	Letter of support signed
SNSF institutional confirmation letter	Requested	UNIGE Vice-Rectorate, standard administrative step
CASU-144 records, primary outcome	**Access held (GESICA)**	Daily, near real time, project agreement and letter in preparation
ED presentations and ICU occupancy	In progress	Extraction in discussion with HUG, letters requested
COVID-19 hospitalisation series (HUG, daily)	Re-request planned	Previously analysed [Orel 2024], validation and calibration series
Pharmacien cantonal, wastewater and pharmacy sales	Contacts established	Extracts to formalise, partly within GESICA
Open public series (surveillance, wastewater, deaths, weather)	**Public**	Freely accessible, no agreement needed

The unresolved feasibility items are the ED/ICU extraction and the project-specific CASU-144 agreement. Each rests on an existing relationship, and neither is claimed as secured yet.

Requested project funds are capped at CHF 250,000 over four years, and my salary is covered separately. Under the 2026 rules, doctoral students and postdocs cannot be employed through Ambizione, but support personnel can.

Item	Rationale
Scientific/technical collaborator, **50% over 48 months**	T1.2 protocol and adjudication support, WP3 harmonisation and evaluation pipeline, reproducibility
Second independent extractor	Contracted M3–M9, required for T1.2's dual-extraction design
Data access, secure storage and processing	Project-specific extractions, secure-environment allocations and runs beyond existing infrastructure
Travel, conferences, incoming visits	Presentation of results, international collaboration
Other eligible direct costs	As justified in the final SNSF budget

Before month 1 and outside the grant, I will finalise project-specific data agreements with HUG and CASU-144, confirm data reuse under the applicable governance and regulatory framework, and harden LiteRev-Evidence before clinical data are connected.

2.4 Schedule and milestones

I will lead WP2 and the confirmatory work in WP3, and lead WP1's protocol and WP4's decision analysis. A 50% scientific/technical collaborator for 48 months will support the extraction benchmark, data harmonisation, forecast-evaluation pipeline and reproducibility. A separately contracted independent extractor will provide the second extraction arm for T1.2 from months 3 to 9.

The work packages can proceed in parallel. WP2 can use weakly informative distributions if extraction error proves unacceptable, the open-data evaluation can proceed without clinical records, and WP4's retrospective decision analysis does not depend on live deployment. The open-data track cannot replace the operational demand test, so access agreements remain a pre-award priority.

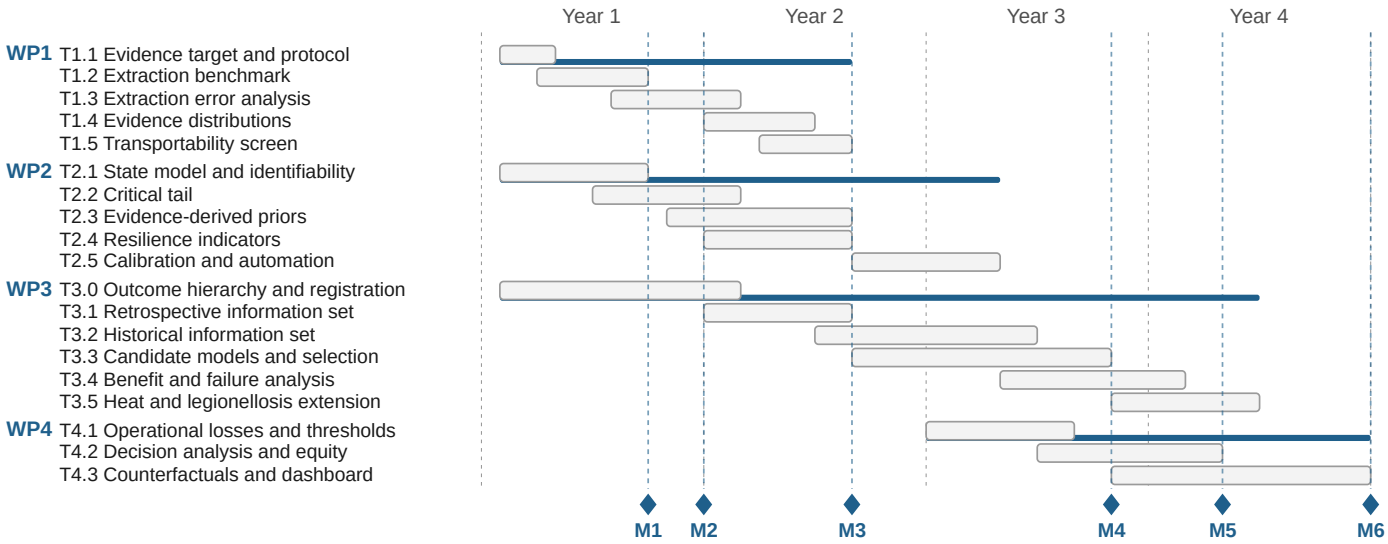


Figure 2. Work packages, task durations and milestones across the 48-month project.

Milestone	Month	Criterion
M1	9	Extraction benchmark released
M2	12	State model passes identifiability criteria, or fallback selected
M3	20	Open-data benchmark running, operational series harmonised (or fallback)
M4	34	Primary early-episode hypothesis tested by the forecast comparison
M5	40	Decision relevance established
M6	48	Analysis across crisis types complete; dashboard for validated demand-based settings where authorised

2.5 Relevance and impact

If the method does not improve forecasts, that is still a useful result. It would show where using published estimates as model inputs adds little value or introduces too much risk, helping modellers and evidence platforms avoid assumptions that have not been tested.

Results will be published in peer-reviewed open-access articles, with preprints where appropriate. The extraction benchmark, forecasting benchmark, evidence-derived input library and code will be released as versioned, citable resources, and findings will be returned to the participating services.

The project will quantify whether external evidence improves forecasts during the first weeks of a crisis, under what conditions it helps, and how uncertain the result remains. For demand-based

applications that pass validation and receive authorisation, the project will provide a registered forecasting configuration and an observation-mode dashboard. Broader deployment across additional crises is a possible post-grant development, not a required project outcome.

2.6 Relevance for personal career development

I have developed the methods, prototype and data relationships that make this project feasible, but much of that work has been carried out within existing programmes. HEADSTART gives me a separate scientific question and a programme that I will lead.

Programme	Objective	My role	Distinction from Ambizione
GESICA	Franco-Swiss crisis intelligence	Contributor	Decision support, no test of evidence-derived priors
GeoAI4EI (Horizon Europe)	European epidemic-intelligence toolbox	Contributor	Infrastructure, no test of borrowed evidence
Legionellosis study (ongoing)	Geneva waterborne outbreak determinants	Data lead	Supplies a contrasting validation extension
Ambizione	When published evidence improves forecasts, and when it misleads	**PI, full-time**	-

HEADSTART is distinct from GESICA and GeoAI4EI, which develop epidemic-intelligence and decision-support capabilities rather than testing published evidence as a forecasting input.

I conceived and built LiteRev with Aziza Merzouki in Prof. Olivia Keiser's group. My current extension, LiteRev-Evidence, adds structured numerical extraction and modelling. It can turn published estimates into parameter distributions, but it has not yet been established whether using those distributions improves forecasts in a new setting. My quantitative-finance work provides methods for rare events, regime changes and stress testing, and my work on LiteRev-Evidence provides the structured evidence pipeline. The research question and the confirmatory analyses for HEADSTART are mine.

Prof. Olivia Keiser led the group in which LiteRev was built and is also involved in the ongoing legionellosis study and GeoAI4EI. She has no scientific role in HEADSTART. Prof. Douglas Teodoro leads GESICA and supports the biomedical-NLP collaboration needed for WP1.

The Institute of Global Health will host the project, but the research programme, hypotheses and analyses will be led by me rather than embedded in an existing group. The institutional and collaboration arrangements will be documented in the required letters. The collaborator and independent extractor support defined technical tasks, and scientific direction remains with the PI.

Where this leads. By month 48, a validated answer and a first-author record, the basis for group leadership or a professorship.

Bibliography

- **[Angelopoulos 2023]** Angelopoulos AN, Bates S. Conformal prediction: a gentle introduction. *Foundations and Trends in Machine Learning* 2023;16(4):494–591.
- **[Barber 2023]** Barber RF, Candès EJ, Ramdas A, Tibshirani RJ. Conformal prediction beyond exchangeability. *Annals of Statistics* 2023;51(2):816–845.
- **[Bareinboim 2016]** Bareinboim E, Pearl J. Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences* 2016;113(27):7345–7352. doi:10.1073/pnas.1510507113

- **[Bonora 2025]** Bonora R, Ticozzi EM, Pregliasco FE, Pagliosa A, Bodina A, Cereda D, Perotti G, Lombardo M, Stirparo G. Telephone calls to emergency medical service as a tool to predict influenza-like illness: a 10-year study. *Public Health* 2025;238:239–244. doi:10.1016/j.puhe.2024.12.021
- **[Brett 2018]** Brett TS, O'Dea EB, Marty É, Miller PB, Park AW, Drake JM, Rohani P. Anticipating epidemic transitions with imperfect data. *PLoS Computational Biology* 2018;14(6):e1006204. doi:10.1371/journal.pcbi.1006204
- **[Coles 2001]** Coles S. *An Introduction to Statistical Modeling of Extreme Values.* Springer, 2001. doi:10.1007/978-1-4471-3675-0
- **[Cook 2023]** Cook JD, Williams DM, Walsh DP, Hefley TJ. Bayesian forecasting of disease spread with little or no local data. *Scientific Reports* 2023;13:8137. doi:10.1038/s41598-023-35177-6
- **[Cramer 2022]** Cramer EY, Ray EL, Lopez VK, et al. (consortium, more than 50 authors). Evaluation of individual and ensemble probabilistic forecasts of COVID-19 mortality in the United States. *Proceedings of the National Academy of Sciences* 2022;119(15):e2113561119. doi:10.1073/pnas.2113561119
- **[Dahabreh 2020]** Dahabreh IJ, Petito LC, Robertson SE, Hernán MA, Steingrimsson JA. Toward causally interpretable meta-analysis: transporting inferences from multiple randomized trials to a new target population. *Epidemiology* 2020;31(3):334–344. doi:10.1097/EDE.0000000000001177
- **[Degtiar 2023]** Degtiar I, Rose S. A review of generalizability and transportability. *Annual Review of Statistics and Its Application* 2023;10:501–524.
- **[Edjinedja 2026]** Edjinedja KL, Larribau R, Orel E, Cossus J, Elfahim O, Yendouname K, Vaussenat F, Teodoro D, Robert-Nicoud S, Barakat O, Desmettre T. Artificial intelligence in emergency medical services for disasters and health emergencies: a systematic review. Submitted, 2026.
- **[Elliott 2014]** Elliott JH, Turner T, Clavisi O, Thomas J, Higgins JPT, Mavergames C, Gruen RL. Living systematic reviews: an emerging opportunity to narrow the evidence-practice gap. *PLoS Medicine* 2014;11(2):e1001603. doi:10.1371/journal.pmed.1001603
- **[Esra 2023]** Esra R, Carstens J, Le Roux S, Mabuto T, Eisenstein M, Keiser O, Orel E, Merzouki A, De Voux L, Maskew M, Sharpey-Schafer K. Validation and improvement of a machine learning model to predict interruptions in antiretroviral treatment in South Africa. *Journal of Acquired Immune Deficiency Syndromes* 2023;92(1):42–49.
- **[Estill 2020]** Estill J, Venkova-Marchevska P, Roelens M, Orel E, Temerev A, Flahault A, Keiser O. Future scenarios for the SARS-CoV-2 epidemic in Switzerland: an age-structured model. *F1000Research* 2020;9.
- **[Farrington 1996]** Farrington CP, Andrews NJ, Beale AD, Catchpole MA. A statistical algorithm for the early detection of outbreaks of infectious disease. *Journal of the Royal Statistical Society Series A* 1996;159(3):547–563.
- **[Gneiting 2007]** Gneiting T, Raftery AE. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 2007;102(477):359–378.
- **[Hamilton 1989]** Hamilton JD. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* 1989;57(2):357–384.

- **[Hobbs 2011]** Hobbs BP, Carlin BP, Mandrekar SJ, Sargent DJ. Hierarchical commensurate and power prior models for adaptive incorporation of historical information in clinical trials. *Biometrics* 2011;67(3):1047–1056.
- **[Höhle 2014]** Höhle M, an der Heiden M. Bayesian nowcasting during the STEC O104:H4 outbreak in Germany, 2011. *Biometrics* 2014;70(4):993–1002. doi:10.1111/biom.12194
- **[Ibrahim 2000]** Ibrahim JG, Chen M-H. Power prior distributions for regression models. *Statistical Science* 2000;15(1):46–60.
- **[Keeling 2008]** Keeling MJ, Rohani P. *Modeling Infectious Diseases in Humans and Animals.* Princeton University Press, 2008.
- **[Lazer 2014]** Lazer D, Kennedy R, King G, Vespignani A. The parable of Google Flu: traps in big data analysis. *Science* 2014;343(6176):1203–1205. doi:10.1126/science.1248506
- **[Le Strat 1999]** Le Strat Y, Carrat F. Monitoring epidemiologic surveillance data using hidden Markov models. *Statistics in Medicine* 1999;18(24):3463–3478.
- **[Lung 2021]** Lung S-CC, Yeh J-CJ, Hwang J-S. Selecting thresholds of heat-warning systems with substantial enhancement of essential population health outcomes for facilitating implementation. *International Journal of Environmental Research and Public Health* 2021;18(18):9506. doi:10.3390/ijerph18189506
- **[McGough 2020]** McGough SF, Johansson MA, Lipsitch M, Menzies NA. Nowcasting by Bayesian smoothing: a flexible, generalizable model for real-time epidemic tracking. *PLoS Computational Biology* 2020;16(4):e1007735. doi:10.1371/journal.pcbi.1007735
- **[Merzouki 2021]** Merzouki A, Estill J, Orel E, Tal K, Keiser O. Clusters of sub-Saharan African countries based on sociobehavioural characteristics and associated HIV incidence. *PeerJ* 2021;9:e10660. doi:10.7717/peerj.10660
- **[Ng'ambi 2026]** Ng'ambi W, Estill J, Keiser O, Merzouki FA, Orel E. Machine learning-based classification of self-reported cardiovascular disease history in Africa using harmonised multi-country WHO STEPS surveys: 2014–2019. Accepted for publication.
- **[Noufaily 2013]** Noufaily A, Enki DG, Farrington P, Garthwaite P, Andrews N, Charlett A. An improved algorithm for outbreak detection in multiple surveillance systems. *Statistics in Medicine* 2013;32(7):1206–1222. doi:10.1002/sim.5595
- **[Nwosu 2021]** Nwosu KD, Fokam J, Wanda F, Mama L, Orel E, Ray N, Meke J, Tassegning A, Takou D, Mimbe E, Stoll B, Guillebert J, Comte E, Keiser O, Ciaffi L. SARS-CoV-2 antibody seroprevalence and associated risk factors in an urban district in Cameroon. *Nature Communications* 2021;12:5851. doi:10.1038/s41467-021-25946-0
- **[O'Regan 2013]** O'Regan SM, Drake JM. Theory of early warning signals of disease emergence and leading indicators of elimination. *Theoretical Ecology* 2013;6:333–357.
- **[Orel 2022]** Orel E, Esra R, Estill J, Thiabaud A, Marchand-Maillet S, Merzouki A, Keiser O. Prediction of HIV status based on socio-behavioural characteristics in East and Southern Africa. *PLoS ONE* 2022;17(3):e0264429. doi:10.1371/journal.pone.0264429
- **[Orel 2023]** Orel E, Ciglenecki I, Thiabaud A, Temerev A, Calmy A, Keiser O, Merzouki A. An automated literature review tool (LiteRev) for streamlining and accelerating research using natural language processing and machine learning. *Journal of Medical Internet Research* 2023;25:e39736. doi:10.2196/39736
- **[Orel 2024]** Orel E, Sobel J, Zanella MC, Iten A, Kaiser L, Keiser O, Vetter P. Clinical severity of Delta and 3 different Omicron sublineages in patients hospitalized because of COVID-19 in a

Swiss tertiary center. *CMI Communications* 2024;1(2):100005. doi:10.1016/j.cmicom.2024.100005

- **[Ragettli 2019]** Ragettli MS, Vicedo-Cabrera AM, Schindler C, Röösli M. Impact of the warm summer 2015 on emergency hospital admissions in Switzerland. *Environmental Health* 2019;18:66. doi:10.1186/s12940-019-0507-1
- **[Ranjbar 2022]** Ranjbar S, Cantoni E, Chavez-Demoulin V, Marra G, Radice R, Jaton-Ogay K. Modelling the extremes of seasonal viruses and hospital congestion: the example of flu in a Swiss hospital. *Journal of the Royal Statistical Society Series C (Applied Statistics)* 2022;71(4):884–905. doi:10.1111/rssc.12559
- **[Rosenkötter 2013]** Rosenkötter N, Ziemann A, Garcia-Castrillo Riesgo L, Gillet JB, Vergeiner G, Krafft T, Brand H. Validity and timeliness of syndromic influenza surveillance during the autumn/winter wave of A(H1N1) influenza 2009: results of emergency medical dispatch, ambulance and emergency department data from three European regions. *BMC Public Health* 2013;13:905. doi:10.1186/1471-2458-13-905
- **[Salmon 2016]** Salmon M, Schumacher D, Höhle M. Monitoring count time series in R: aberration detection in public health surveillance. *Journal of Statistical Software* 2016;70(10):1–35. doi:10.18637/jss.v070.i10
- **[Scheffer 2009]** Scheffer M, Bascompte J, Brock WA, Brovkin V, Carpenter SR, Dakos V, Held H, van Nes EH, Rietkerk M, Sugihara G. Early-warning signals for critical transitions. *Nature* 2009;461:53–59. doi:10.1038/nature08227
- **[Schmidli 2014]** Schmidli H, Gsteiger S, Roychoudhury S, O'Hagan A, Spiegelhalter D, Neuenschwander B. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics* 2014;70(4):1023–1032. doi:10.1111/biom.12242
- **[Schulte 2024]** Schulte F, Röösli M, Ragettli MS. Risk, attributable fraction and attributable number of cause-specific heat-related emergency hospital admissions in Switzerland. *International Journal of Public Health* 2024;69:1607349. doi:10.3389/ijph.2024.1607349
- **[Shankar 2026]** Shankar R, Lim A, Qian X. Performance of large language models in data extraction for evidence synthesis: a systematic review. *Journal of Biomedical Informatics* 2026;181:105086. doi:10.1016/j.jbi.2026.105086
- **[Sherratt 2023]** Sherratt K, Gruson H, Grah R, et al. (consortium, more than 50 authors). Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations. *eLife* 2023;12:e81916. doi:10.7554/eLife.81916
- **[Southall 2021]** Southall E, Brett TS, Tildesley MJ, Dyson L. Early warning signals of infectious disease transitions: a review. *Journal of the Royal Society Interface* 2021;18(182):20210555. doi:10.1098/rsif.2021.0555
- **[Vickers 2006]** Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making* 2006;26(6):565–574.
- **[Wargon 2009]** Wargon M, Guidet B, Hoang TD, Hejblum G. A systematic review of models for forecasting the number of emergency department visits. *Emergency Medicine Journal* 2009;26(6):395–399. doi:10.1136/emj.2008.062380
- **[Watkins 2009]** Watkins RE, Eagleson S, Veenendaal B, Wright G, Plant AJ. Disease surveillance using a hidden Markov model. *BMC Medical Informatics and Decision Making* 2009;9:39. doi:10.1186/1472-6947-9-39

- **[Zheng 2015]** Zheng XY, Ding H, Jiang LN, Chen SW, Zheng JP, Qiu M, Zhou YX, Chen Q, Guan WJ. Association between air pollutants and asthma emergency room visits and hospital admissions in time series studies: a systematic review and meta-analysis. *PLoS ONE* 2015;10(9):e0138146. doi:10.1371/journal.pone.0138146